The Proposed Skew Constant Characterizes Inner Product Spaces

Math discovery packet

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1 Source Problem

The source paper is J. Qi, Q. Li, Z. Yin, Q. Liu, J. Bi, Y. Fu, and Y. Li, *Generalized geometric constants related to Birkhoff orthogonality in Banach spaces*, arXiv:2602.13974. In its final open questions, Problem 2 asks whether one can improve the proof of the Ficken inner-product criterion by introducing the constant

$$F(X) = \sup_{t>0} \left\{ \frac{\|x+ty\| - \|tx+y\|}{2} : x, y \in S_X \right\}.$$

At the end of the article, we would like to pose some open questions:

Problem 1:

Due to the properties of Birkhoff orthogonality, it is an interesting topic to incorporate the classical von Neumann-Jordan constant into the framework of Birkhoff orthogonality and redefine it as follows:

$$C_{\mathrm{NJ}}(X) = \sup \left\{ \frac{\|x+y\|^2 + \|x-y\|^2}{2\left(\|x\|^2 + \|y\|^2\right)} : x, y \in X,\ (x,y) \neq (0,0),\ x \perp_B y \right\}.$$

Of course, the estimation of related inequalities also presents considerable difficulties. Furthermore, it is also natural to incorporate the p-th von Neumann-Jordan constant [39] into Birkhoff orthogonality. For related results on the p-th Birkhoff orthogonality, we refer the reader to [40].

Problem 2:

Remark 4 motivates us to investigate whether we can improve the proof of the well-known Lemma 6 by defining the constant as follows: set

$$F(X) = \sup_{t>0} \left\{ \frac{\|x+ty\| - \|tx+y\|}{2} : x, y \in S_X \right\}.$$

Figure 1: The source paper's final open questions, including Problem 2.

2 Result

Theorem 1. *Let X be a real normed space with* $\dim X \geq 2$. *Then*

$$0 \leq F(X) \leq 1.$$

Moreover,

$$F(X) = 0 \iff X \text{ is an inner product space.}$$

3 Idea of the Proof

The expression defining $F(X)$ is skew in the ordered pair (x, y): interchanging x and y changes its sign. Hence vanishing of the supremum is much stronger than it may first look. It forces equality $\|x + ty\| = \|tx + y\|$ for every unit pair and every $t > 0$. Replacing y by $-y$ supplies the negative values of t, so the usual Ficken criterion applies.

4 Proof

First, $F(X) \geq 0$, since taking $x = y \in S_X$ gives value 0. For the upper bound, for $x, y \in S_X$ and $t > 0$,

$$\|x + ty\| \leq 1 + t, \qquad \|tx + y\| \geq |t - 1|.$$

Therefore

$$\frac{\|x + ty\| - \|tx + y\|}{2} \leq \frac{1 + t - |t - 1|}{2} = \min\{1, t\} \leq 1.$$

Taking the supremum gives $F(X) \leq 1$.

If X is an inner product space and $x, y \in S_X$, then

$$\|x + ty\|^2 = 1 + t^2 + 2t\langle x, y\rangle = \|tx + y\|^2$$

for all real t. Hence $F(X) = 0$.

Conversely, assume $F(X) = 0$. Fix $x, y \in S_X$ and $t > 0$. The definition of F gives

$$\|x + ty\| - \|tx + y\| \leq 0.$$

Applying the same inequality to the pair (y, x) gives

$$\|y + tx\| - \|ty + x\| \leq 0,$$

or equivalently $\|tx + y\| \leq \|x + ty\|$. Thus

$$\|x + ty\| = \|tx + y\| \qquad (x, y \in S_X,\ t > 0).$$

For $t < 0$, write $t = -s$ with $s > 0$ and apply the positive-s equality to the pair $(x, -y)$:

$$\|x - sy\| = \|sx - y\| = \| - sx + y\|.$$

The case $t = 0$ is trivial. By homogeneity, for every pair of nonzero vectors u, v with $\|u\| = \|v\|$, one has

$$\|tu + v\| = \|u + tv\| \qquad \text{for every } t \in \mathbb{R}.$$

This is exactly Ficken's characterization of inner product spaces, quoted as Lemma 6 in the source paper. Hence X is an inner product space.

5 Answer to Problem 2

Problem 2 has an affirmative answer in the following precise sense: the proposed constant $F(X)$ improves the skew-equality criterion to a scalar geometric invariant, and its zero set is exactly the class of inner product spaces.

6 References

- J. Qi, Q. Li, Z. Yin, Q. Liu, J. Bi, Y. Fu, and Y. Li, *Generalized geometric constants related to Birkhoff orthogonality in Banach spaces*, arXiv:2602.13974.

- F. A. Ficken, *Note on the existence of scalar products in normed linear spaces*, Ann. of Math. (2) 45 (1944), 362–366.